



LLM meta-judge

A

Is the generated caption faithful to the speech metadata?



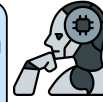
Hey, compare the speaking-style metadata for a ground-truth utterance and an audio-language model's generated caption, think step-by-step about stylistic similarities (speech style, intention, prosody), and output a similarity score from 1–10 (1 = lowest, 10 = highest).

speech_metadata.json



Generated caption : A female American speaker with a high-pitched, shrill voice ...

Hmmmm, seems like you got the gender and speaking rate right, but the pitch characteristics and accent are incorrect. I score this a **7/10**.



LLM hallucination-judge

B

Does the generated caption have hallucinations?



Hey, compare the ground-truth audio metadata and a generated caption, think step-by-step about which caption details are factually supported vs invented or contradictory, and then rate hallucinations on a 0–10 scale (0 = none, 10 = many/severe) and assign an overall severity level (Low/Medium/High).

speech_metadata.json



Generated caption : An Indian male speaker with a sing-song, deep voice speaks in a sad..

I notice that the metadata does not have any indications of a sing-song quality to the voice. However, the other prosodic indicators are on point. This is a **low hallucination** case, I rate it a **3.5/10**



LLM instruction-following judge

C

Does the audio-language model follow instructions?

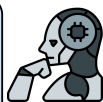


Hey, compare the given instruction and the model's generated caption, identify the key requirements, check how well each one is followed (including format, style, focus, and constraints), and then rate instruction-following as Success/Partial Success/Failure plus an accuracy score from 1–10.

Style Instruction : Generate a speaking-style caption describing the speaker's inherent voice qualities, including timbre, tone ...

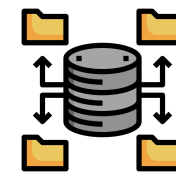
Generated caption : A British male speaker with bass voice is laughing heartily ...

Although the inherent qualities such as gender, accent, tone and timbre are present, the caption also include the situational emotion. This is a **SUCCESS**, however I give it an instruction following score of **6/10**.



Generalization to unseen speech datasets and unseen prompts

D



Does the speaking style captioning model generalize to new, unseen datasets?



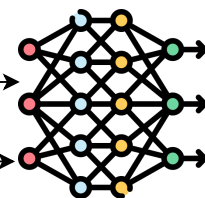
Can the speaking style captioning model understand and respond to unseen style instructions?

Speaker-style consistency analysis

Do the speaker-specific style captions stay consistent across multiple utterances of the same speaker?

E

Speaker Verification



Speaking style caption A



Speaking style caption B

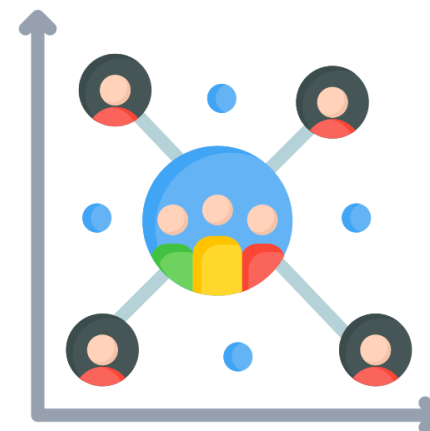
Speaking style caption A
Speaking style caption B

Speaker verification system

Same speaker?
Yes/No

Do speaking style caption A and speaking style caption B belong to the same speaker or different speakers?

Speaker-caption clustering



Do the speaking style captions of the same speakers cluster together in an embedding space?